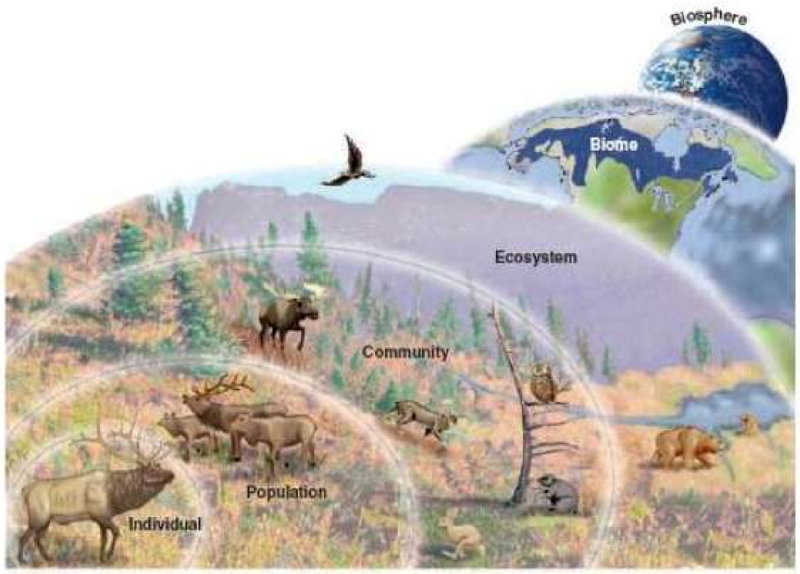
ESc 301.03

The Environmental Dimension

Populations & Communities

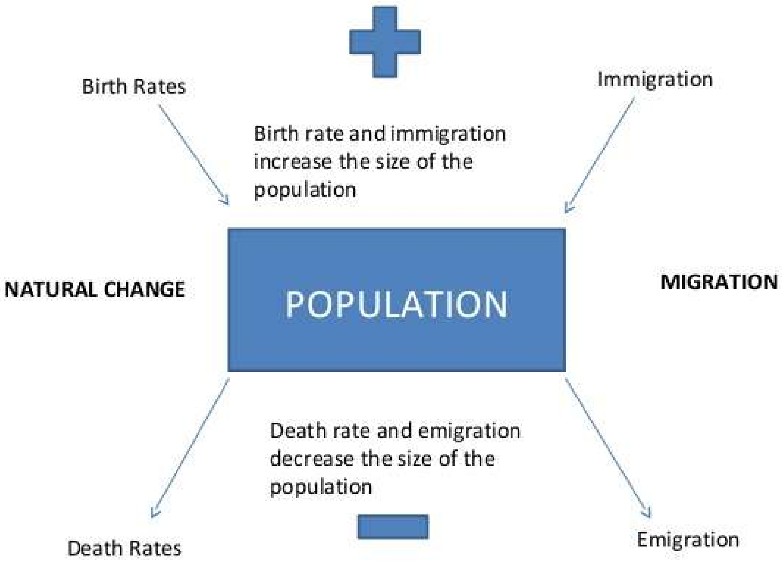


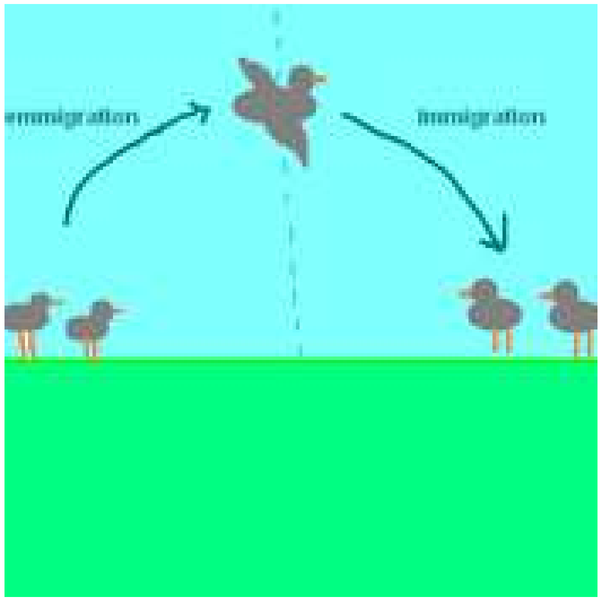
Assoc. Prof. Dr. Başak Güven

Changes in a Population

Affected by 3 factors

- Births
- Deaths
- Migration
  - Immigration
  - Emigration





Population Growth

Factors affecting

Biotic potential

Environmental resistance

Density-independent factors

Density-dependent factors

Survivorship

Age structure

A growth rate example:

- 10.000 birds in a population
- 1500 births and 500 deaths per year
- 1500/10.000 - 500/10.000 = 0.10 or 10 %
- Expressed by saying there is a 10 % increase per bird per year

Biotic Potential and Environmental Resistance

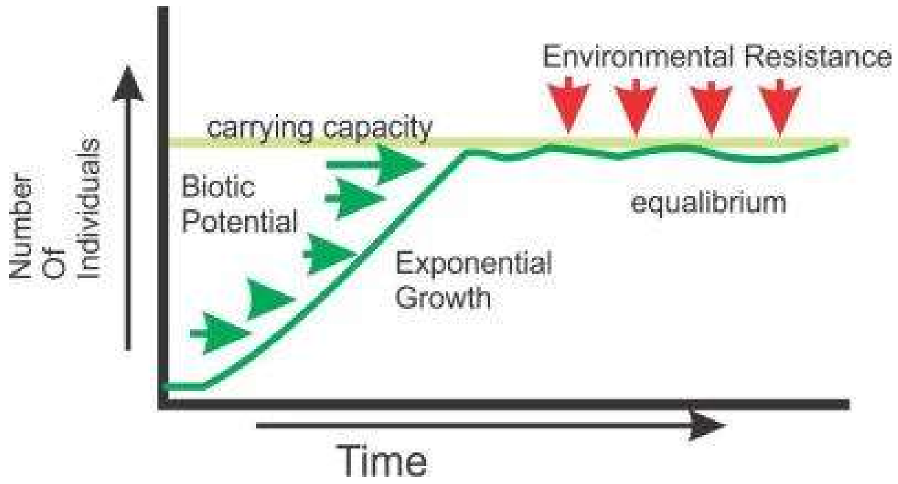
Biotic potential

is the maximum reproductive capacity of a population if resources are unlimited.

Biotic	Abiotic
• Other organisms, so:	Air (O_2, CO_2, N_2, etc)
• Competition	Water
• Predation	Light
• Symbiosis	Wind
– Mutualism	Soil
– Parasitism	pH
• Disease agents	Temperature
	Salinity
	Humidity
	Inorganic nutrients (N, P)
	Etc.

Environmental resistance

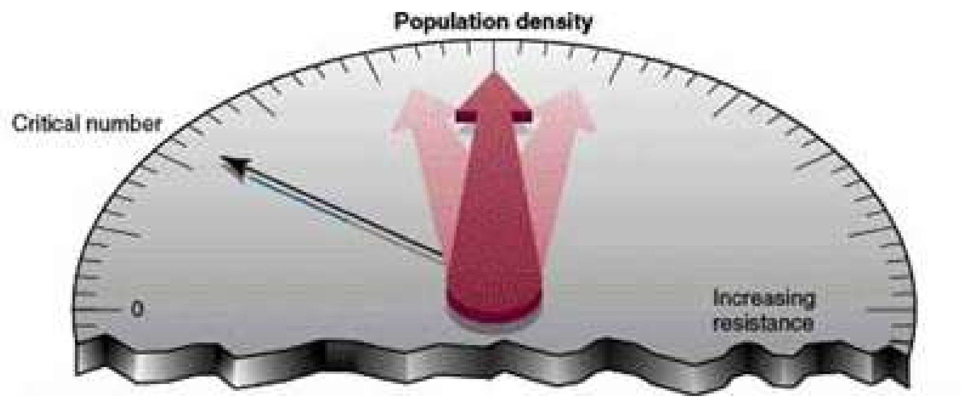
is the combination of biotic and abiotic factors that may limit population increase (predators, competitors, disease, adverse weather, limited food etc.).



<u>Factors of environmental resistance are either:</u>

- **Density-independent:** Effect does not vary with population density (e.g. adverse weather)
- **Density-dependent:** Effect varies with population density (e.g. infectious disease)

Biotic Potential and Environmental Resistance



Biotic potential is dependent upon:

- Reproductive rate
- Ability to migrate
- Ability to invade new habitats
- Defense mechanisms
- Ability to cope with adverse conditions

Environmental Resistance is dependent upon:

- Lack of food for nutrients
- Lack of water
- Lack of suitable habitat
- Adverse weather conditions
- Predators
- Diseases
- Parasites
- Competitors

How Do Populations Grow?

Idealized models describe two kinds of population growth.

Exponential Growth
(J-Shaped)

A **J-shaped growth curve**, described by a typical exponential growth equation.

$$G = r \times N$$

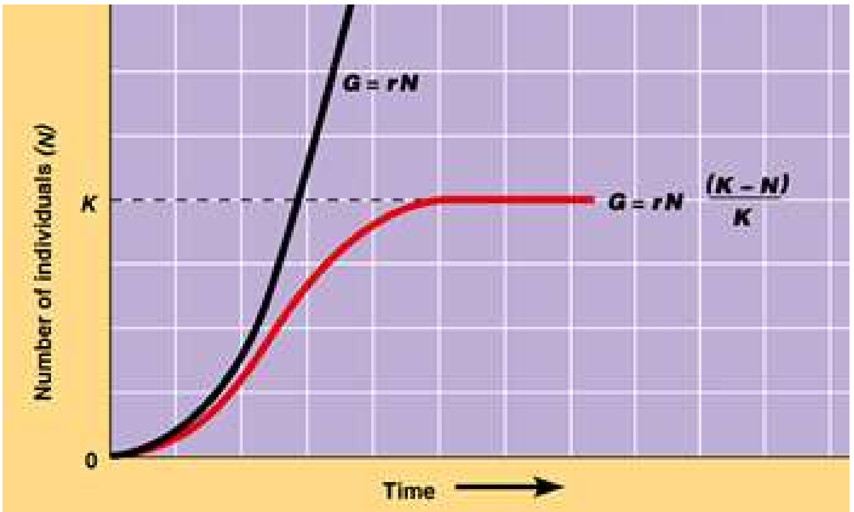
G: population growth rate
r: intrinsic rate of increase, or an organism's maximum capacity to reproduce
N: population size

Logistic Growth
(S-Shaped)

It is slowed by population-limiting factors. **Carrying capacity (K)** is the maximum population size that an environment can support.

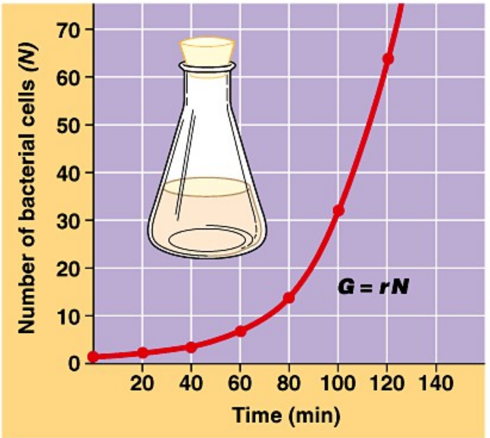
$$G = \frac{\Delta N}{\Delta t} = \mathrm{r} \times N \times \left(\frac{K - N}{K}\right)$$

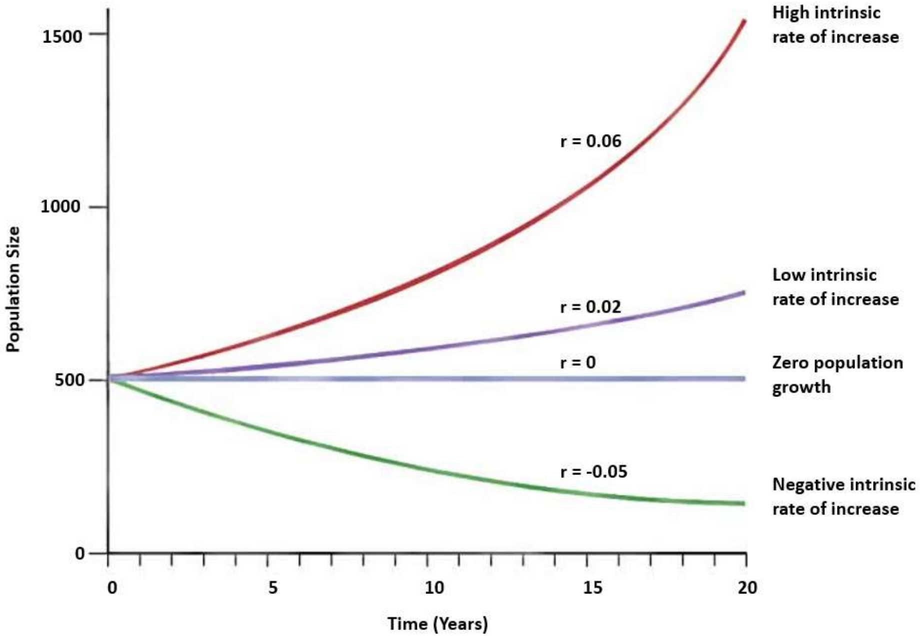
(K - N)/K accounts for the leveling off the curve.



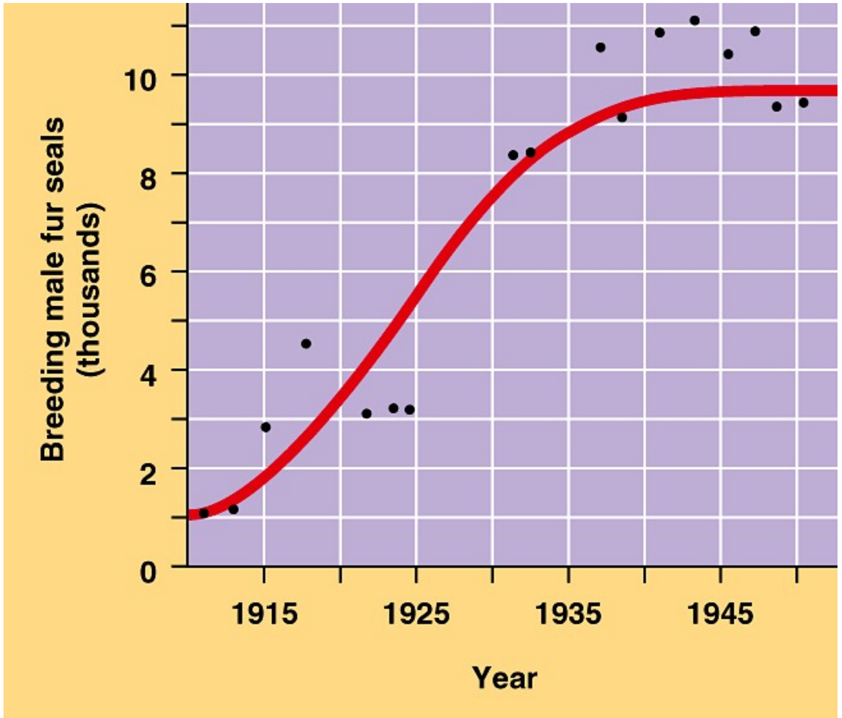
Exponential Growth
(J Shaped)

Time	Number of Cells	
0 minutes	1	$= 2^0$
20	2	$= 2^1$
40	4	$= 2^2$
60	8	$= 2^3$
80	16	$= 2^4$
100	32	$= 2^5$
120 (= 2 hours)	64	$= 2^6$
3 hours	512	$= 2^9$
4 hours	4,096	$= 2^{12}$
8 hours	16,777,216	$= 2^{24}$
12 hours	68,719,476,736	$= 2^{36}$

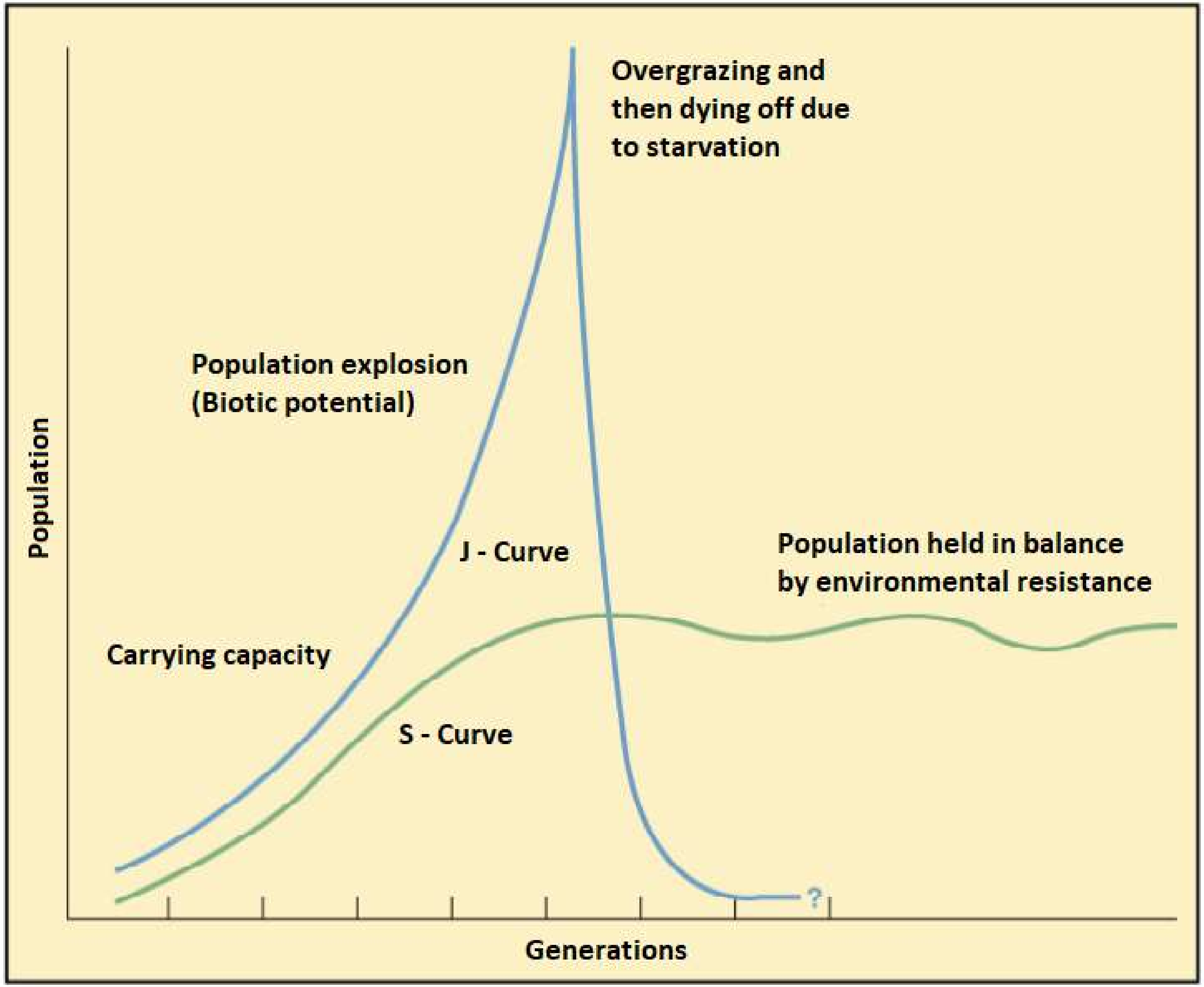




Logistic Growth
(S Shaped)

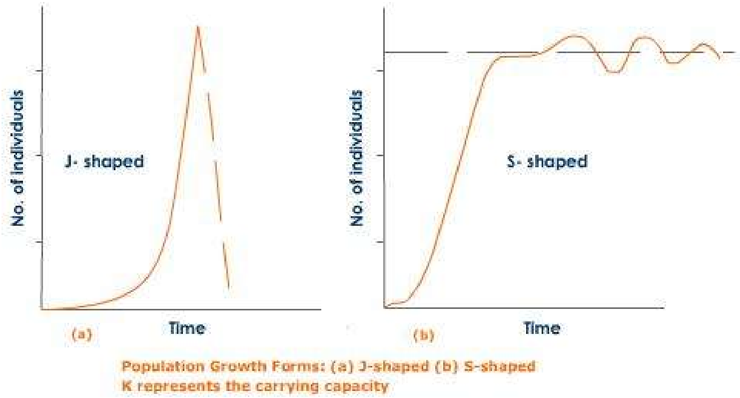


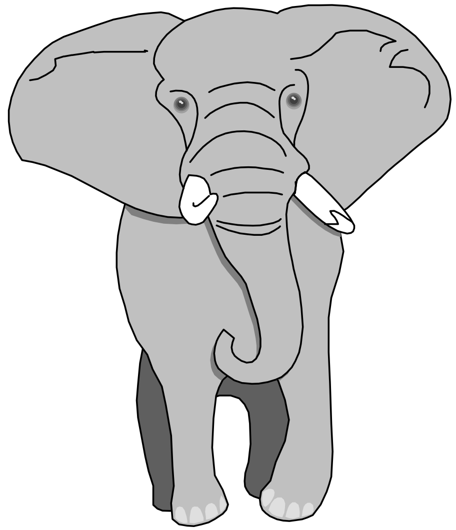
Population Growth Curves



Biotic potential: The maximum reproductive capacity of a population if resources are unlimited.

Population Growth Curves: Reproductive Strategies





Many offspring with low parental care

J-Shaped growth curve

Few offspring with high parental care

S-Shaped growth curve

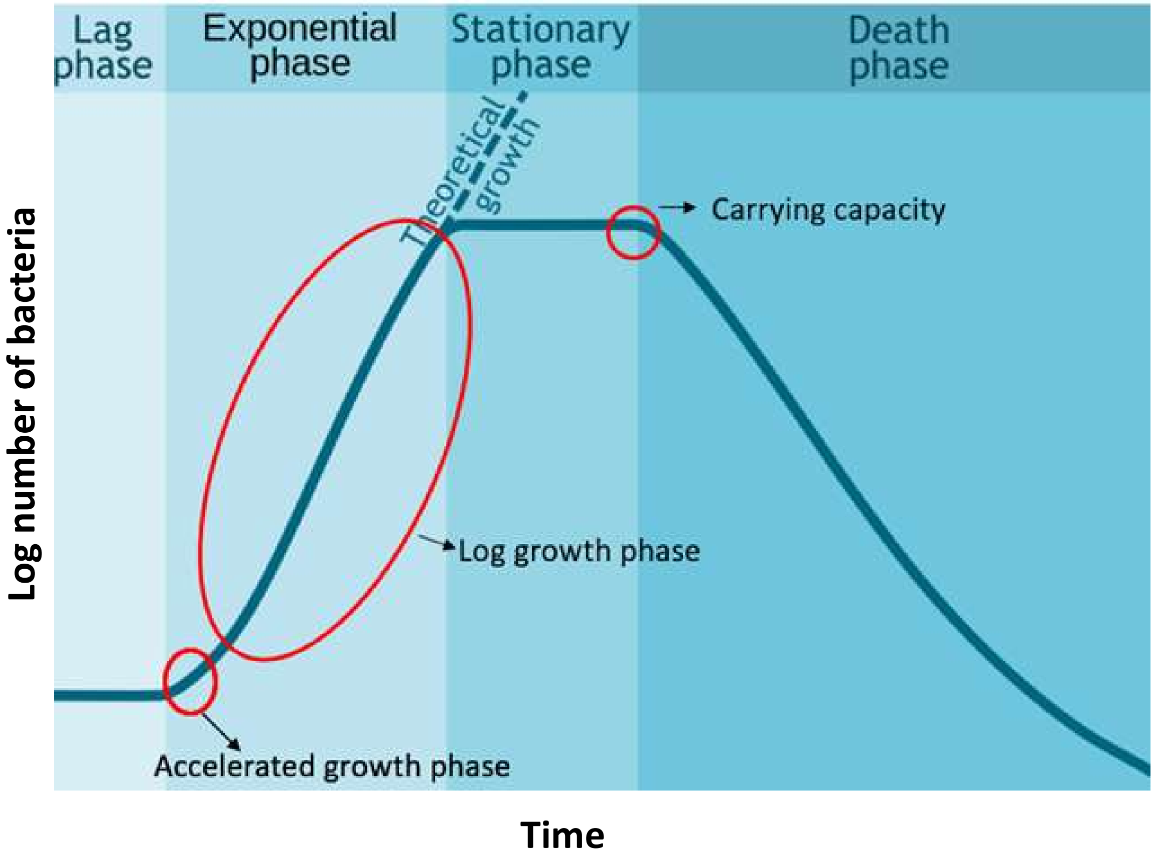
An Example: Bacterial Population Growth

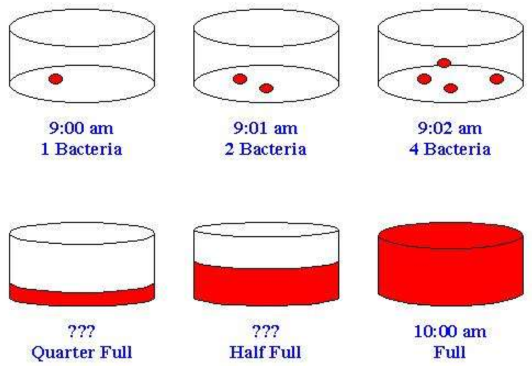
Initially nothing appears to happen.

Bacteria must adjust to their new environment begin to synthesize new protoplasm.

At the end of the lag phase the bacteria begin to divide. Since all of the organisms do not divide at the same time, population increase is gradual. This phase is called as the accelerated growth phase.

At the end of accelerated growth phase, the population of organisms is large enough and the differences in generation time are small enough that the cells appear to divide at a regular rate.



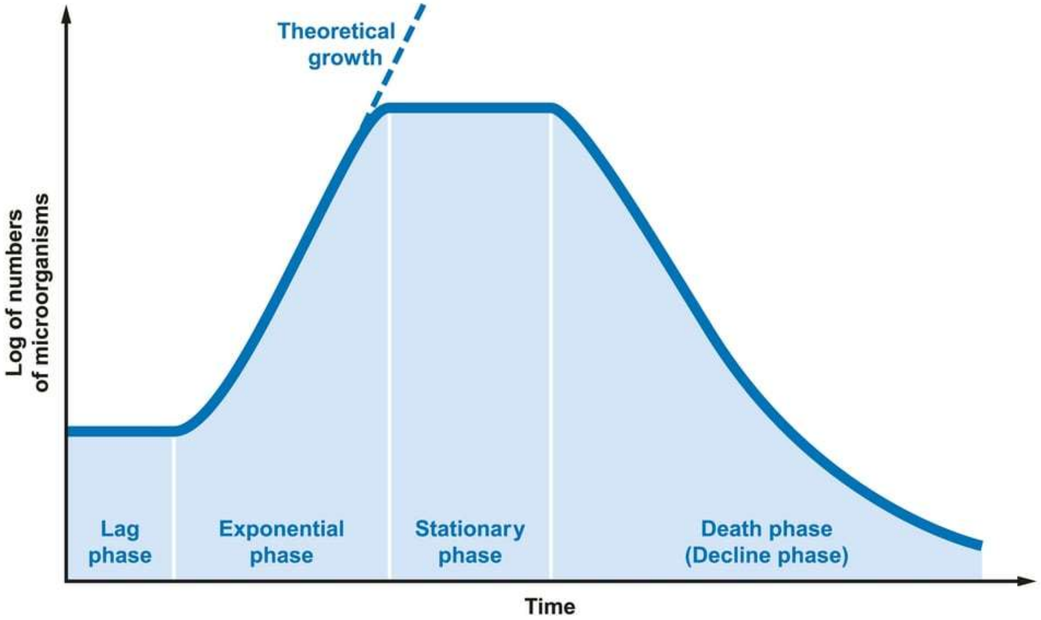


Because reproduction is by binary fission, the increase in population follows in a geometric progression:

$$1 \longrightarrow 2 \longrightarrow 4 \longrightarrow 8 \longrightarrow 16 \longrightarrow 32$$

The population of bacteria (P) after the n^{th} generation is given by the following expression:

$$P = P_0(2)^n$$



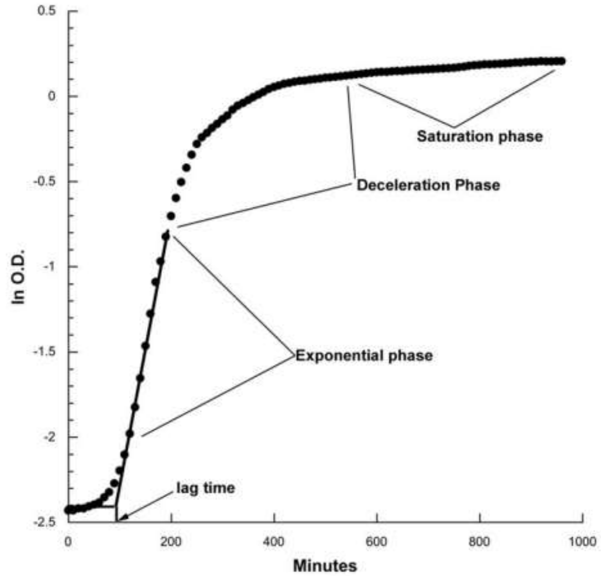
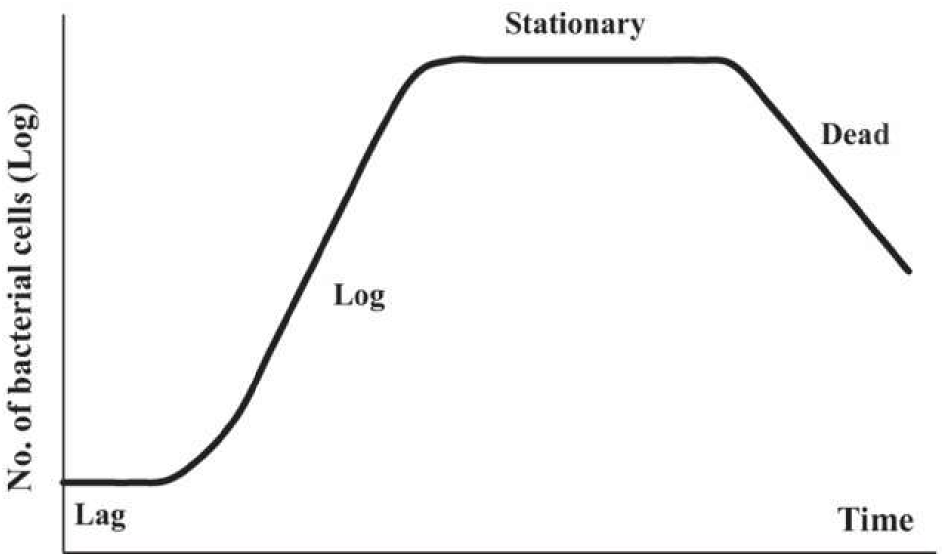


FIG. 1. Typical bacterial growth curve: ln OD is plotted versus time. The points that fall along the straight line represent the exponential growth phase. Here, the lag time is shown as the time to enter exponential phase and really includes the time during which growth accelerates.

https://academic.oup.com/mbe/article/31/1/232/1049132



https://www.nature.com/articles/srep15159

Graphic representation of typical bacterial growth curve in culture medium. When bacteria are introduced into the fresh medium in a closed system, like a test tube, the population of cells always exhibits growth dynamics as follows. Lag phase: bacteria initially adjust to the new environment, so they appear not able to replicate but the cells might grow in volume; Log (exponential) phase: cells start dividing regularly by the process of binary fission. The culture reaches the maximum growth rate, which could be estimated by generation time or doubling time, i.e., the time per generation; Stationary phase: the number of cells undergoing division seems to be equal to that being dead due to the exhaustion of nutrients. Dead phase: bacteria lose the ability to divide and the number of dead cells exceeds that of live cells.

An Example: Human Population Growth

Predicting the dynamics of human populations is important because;

- It is the basis for the determination of design capacity for municipal water and wastewater treatment systems and for water reservoirs.
- It is important in the development of resource and pollutant management plans.

$$P(t) = P_0 e^{rt}$$

population at time, *t*

population at time, 0

rate of growth

time

$$r = b - d + i - m$$

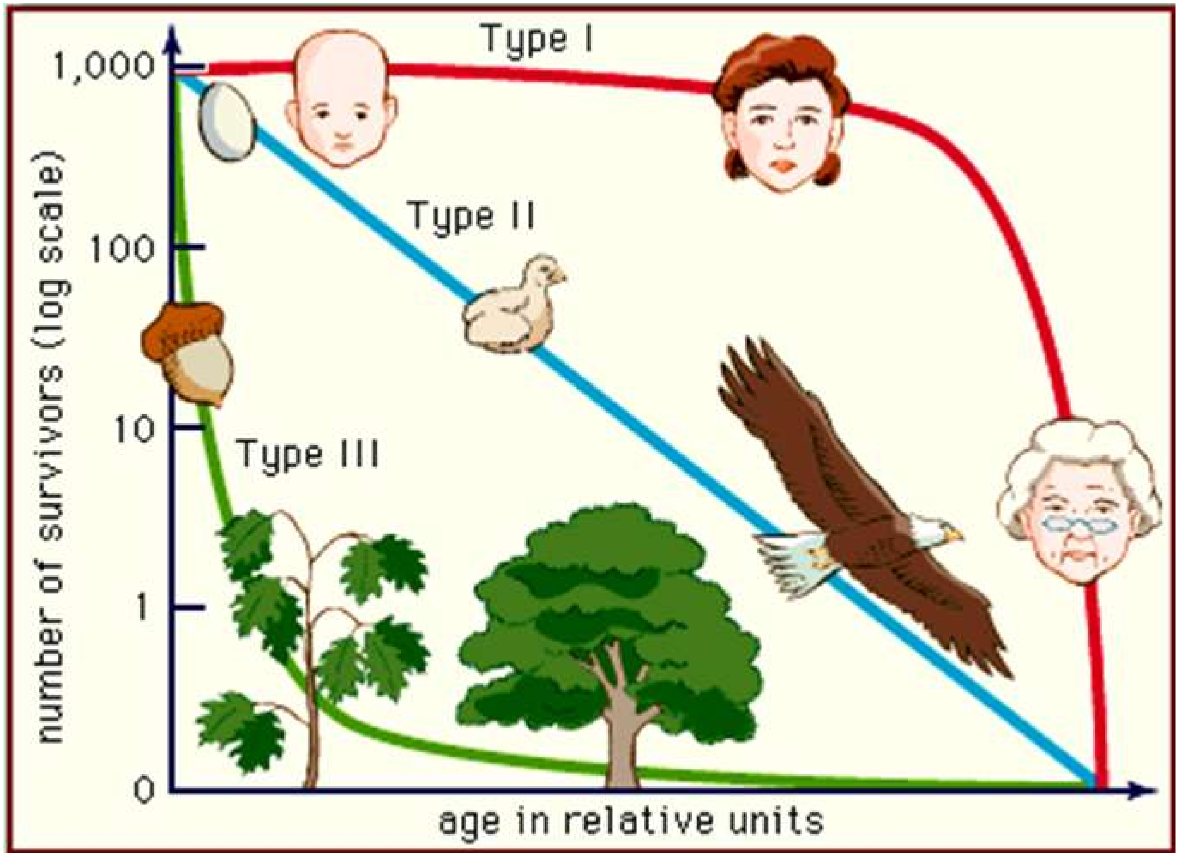
birth rate   death rate   immigration rate   emigration rate

Survivorship

- ❖ Mirrors mortality

- ❖ Expressed in survivorship curves
  - Plots surviving individuals at different age groups

- ❖ Three types of survivorship curves
  - Late loss (Type I)
  - Constant loss (Type II)
  - Early loss (Type III)



The Concept of Habitat & Niche

Habitat: The place where an organism lives.

Examples: A lion's habitat is a savanna. A monkey's habitat is a rain forest. A cactus's habitat is in the desert.

Niche: An organism's way of life; occupation.

Example: A lion's niche includes where and how it finds shelter and food, when it reproduces, how it relates to other animals.

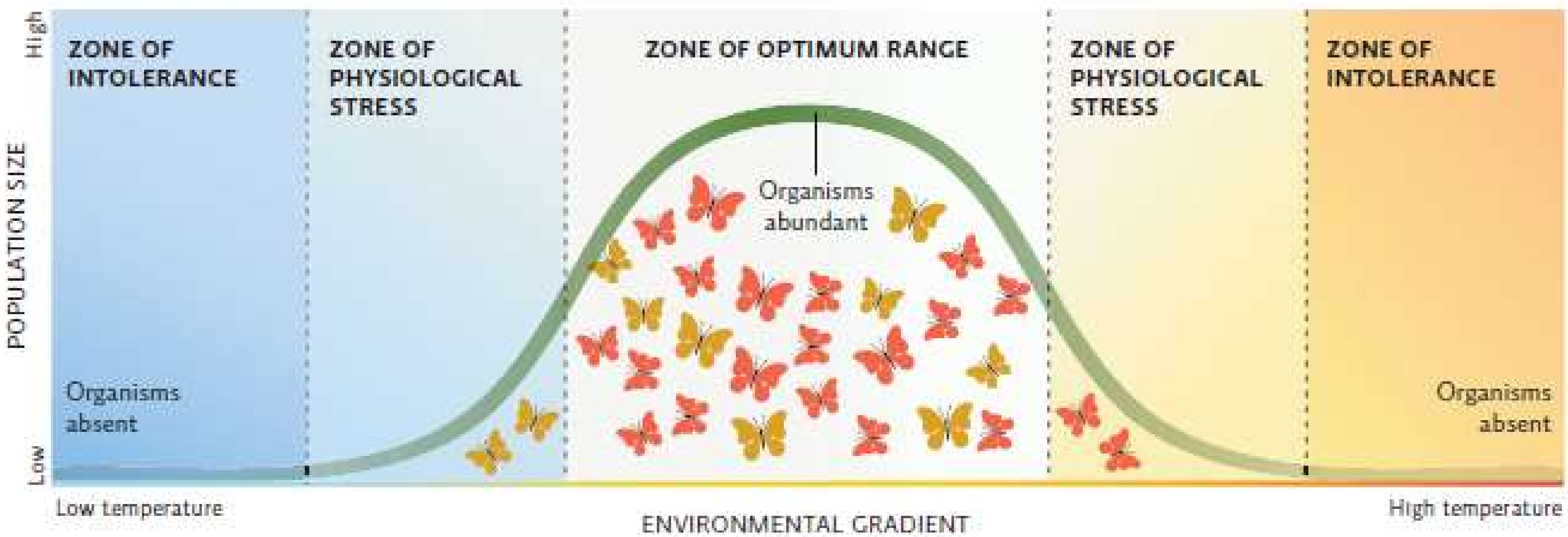
Habitat	Niche
The habitat is the place where an organism lives out its life: i.e. It is where the organism finds food, shelter and mates.	*A niche is its role in the community and how it interacts with the environment: i.e. How it obtains food, mates and protection from predators.*

The Concept of Niche

Habitat niche: The place where an organism lives.

Trophic or food niche: How the organism feeds

Multidimensional niche: Under which conditions the organism lives: pH, T, oxygen level etc.

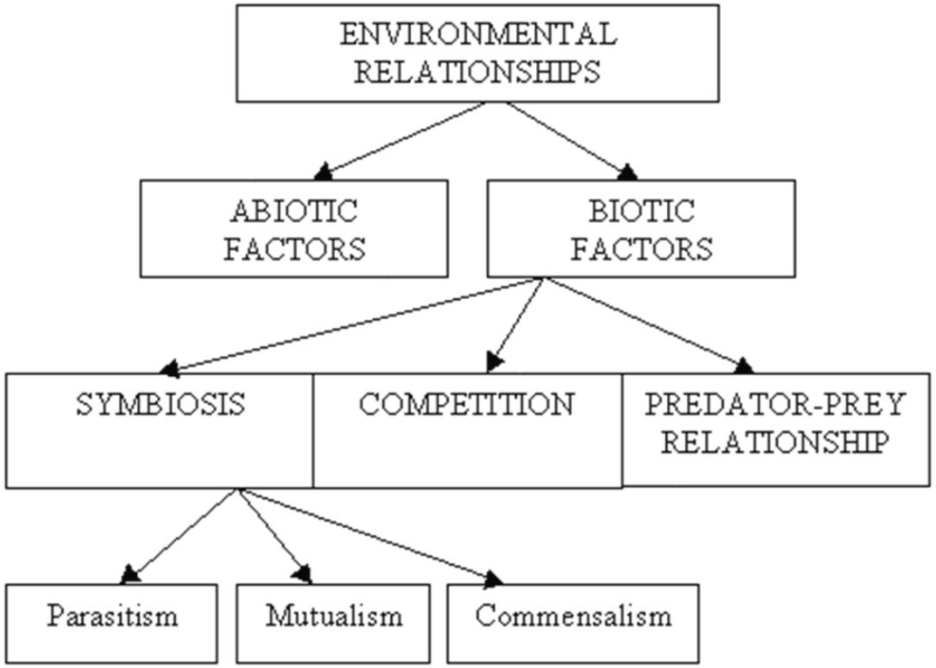


↑ Populations have a range of tolerance for any given environmental factor (such as temperature). Every species has an upper and a lower limit beyond which it cannot survive (in this example, temperatures that are too cool or too hot). Most individuals, like the butterflies in this population, can be found around the optimum temperature, though what is "optimum" for each individual may differ slightly because of genetic variability. Some individuals may find themselves in areas of the habitat that are warmer or colder than the optimum. They can tolerate these conditions but may be physiologically stressed and not grow well or successfully breed. Genetic differences that allow some individuals to tolerate or even thrive at the edges of the population's tolerance offer the population a chance to adapt to changing conditions (such as a warmer climate) if needed. The more narrow the range of tolerance and the less genetically diverse the population, the less likely it will survive a change in conditions.

Communities

A community is a collection of populations of all the organisms, which occur together in a given place and time. Community ecology is the study of the interactions between these organisms, and the interactions between the organisms and their environment.

Species richness, species interactions, trophic structure, succession etc. are important topics in community ecology.



Mutualism: + and +
Both species benefit by the interaction between the two species.

Commensalism: + and 0
One species benefits from the interaction and the other is unaffected

Predation, parasitism: + and –
One species benefits from the interaction and the other is adversely affected.

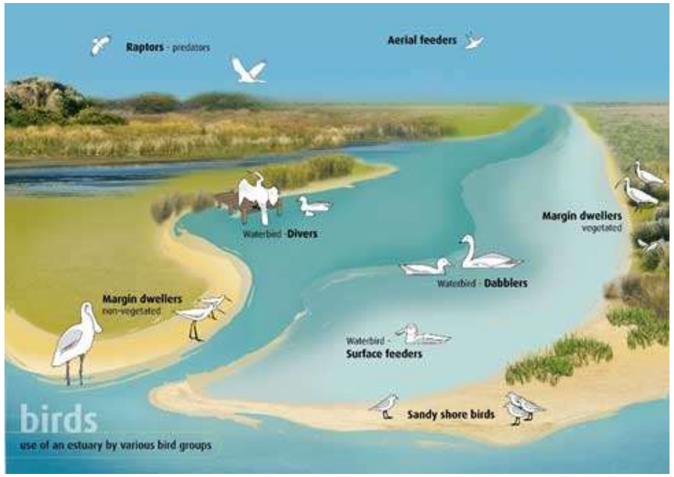
Competition: - and –
Both species are adversely affected by the interaction.

Species Diversity: Ecotone and Edge Effect

An ecotone is a **zone of junction** or a **transition area** between two biomes [diverse ecosystems]. It is where two communities meet and integrate.

Mangrove forests represent an ecotone between marine and terrestrial ecosystem.

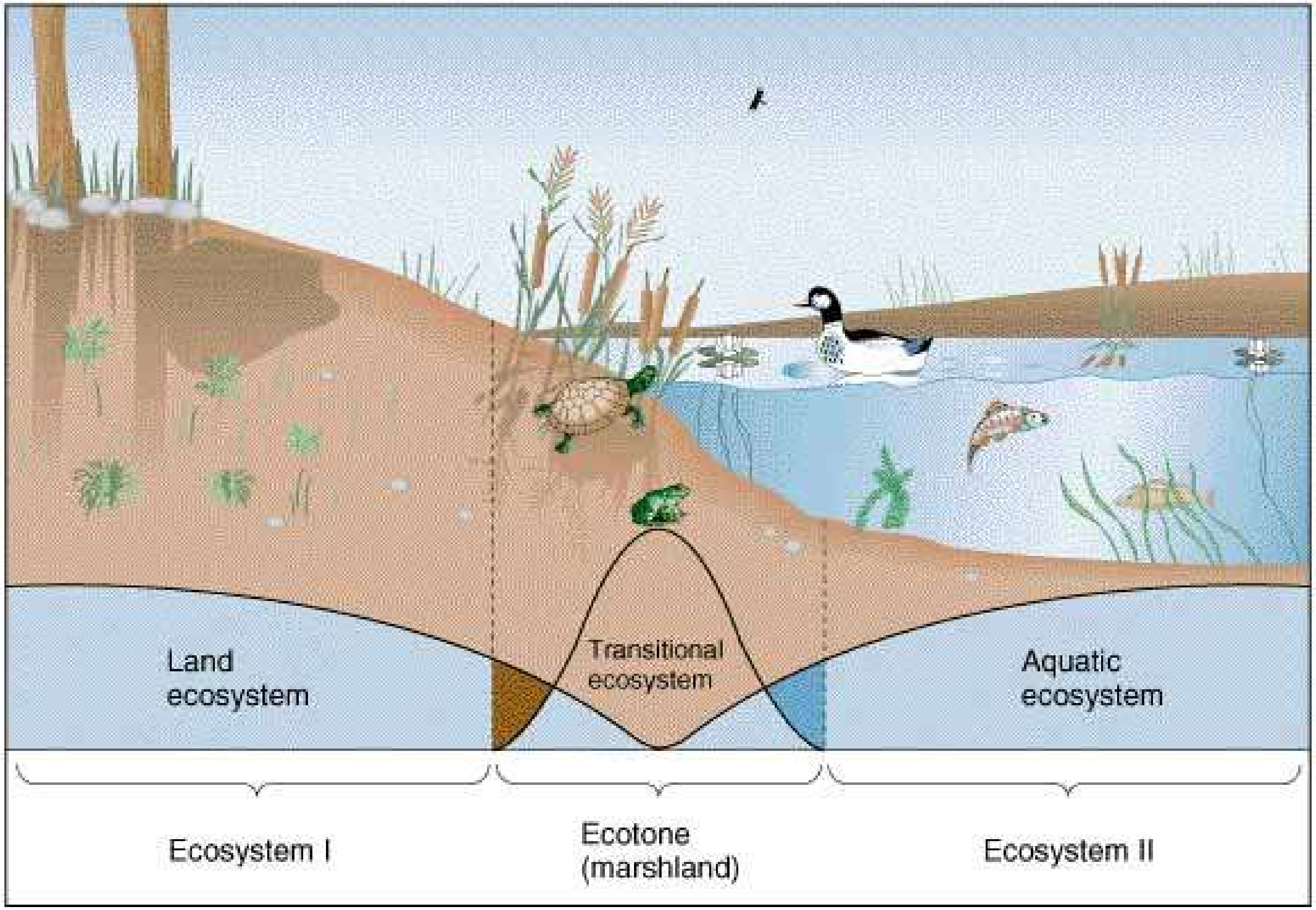
Estuary
(between freshwater and salt water)



Grassland
(between forest and desert)

River bank or **marsh land**
(between dry and wet).

Species Diversity: Ecotone and Edge Effect



In ecology, edge effects refer to the changes in population or community structures that occur at the boundary of two habitats (ecotone).

Succession and Disturbance

Ecological succession is the process by which the structure of a biological community evolves over time.

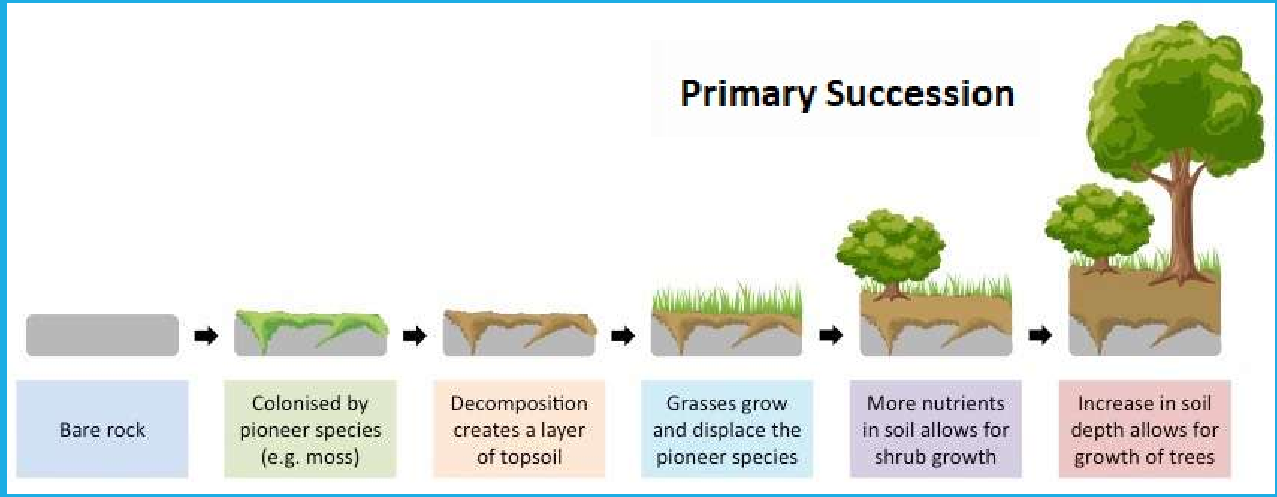
Primary: No previous biotic community

Secondary: Previously occupied by a community

Aquatic: Transition from pond or lake to terrestrial community

Disturbances are events such as floods, fire, droughts, overgrazing, and human activity that damage communities, remove organisms from them, and alter resource availability.

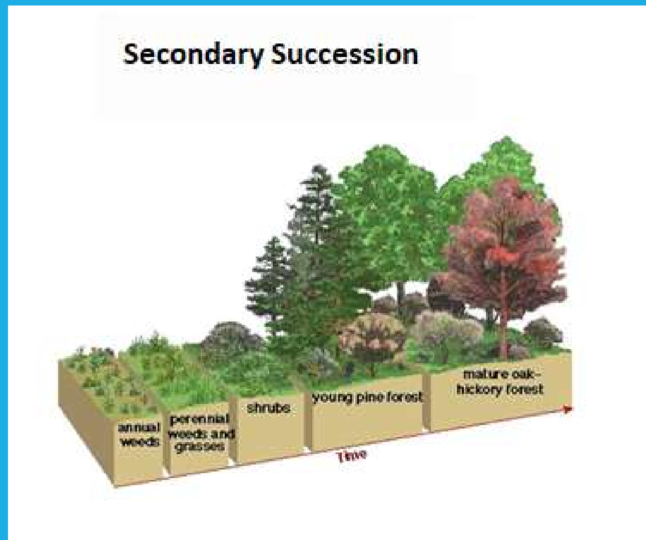
> ***Ecological succession*** *is the term used to describe how an ecosystem gradually changes in species composition and community structure over time after a disturbance has occurred.*



Primary succession is the type of succession that occurs when an area experiences a disturbance so severe that none of the original species survive.

During the process of primary succession, the disturbance causes the exposure of bare rock and the ecosystem is rebuilt from scratch due to the lack of organisms that remain. Primary succession is often a slow process because it requires many steps to convert bare rock to a functional ecosystem. The process normally starts with the invasion of **lichens** and **mosses**, which create soil, and then the process moves on to allow for the establishment of larger vegetation.

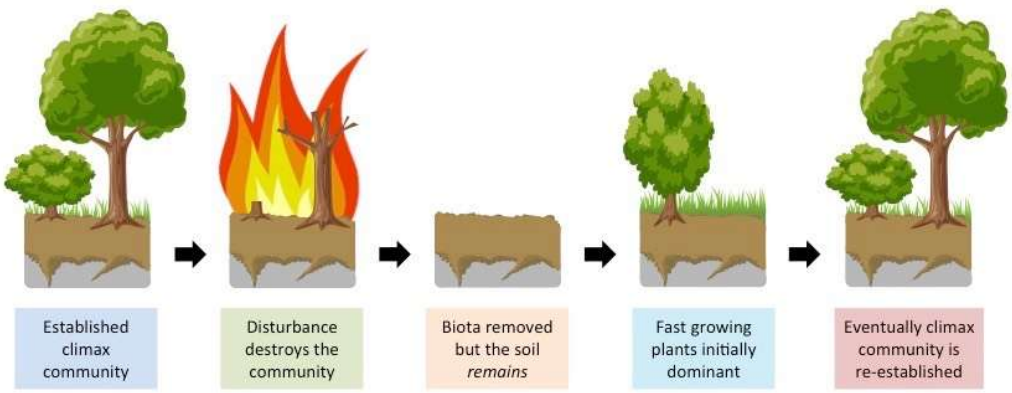
In contrast, **secondary succession** is the type of succession that occurs when an area experiences a disturbance that alters the existing ecosystem but does not destroy all of the original species. During secondary succession, the soil and species that remain after the disturbance are used as the building blocks that help facilitate the recovery of the ecosystem.

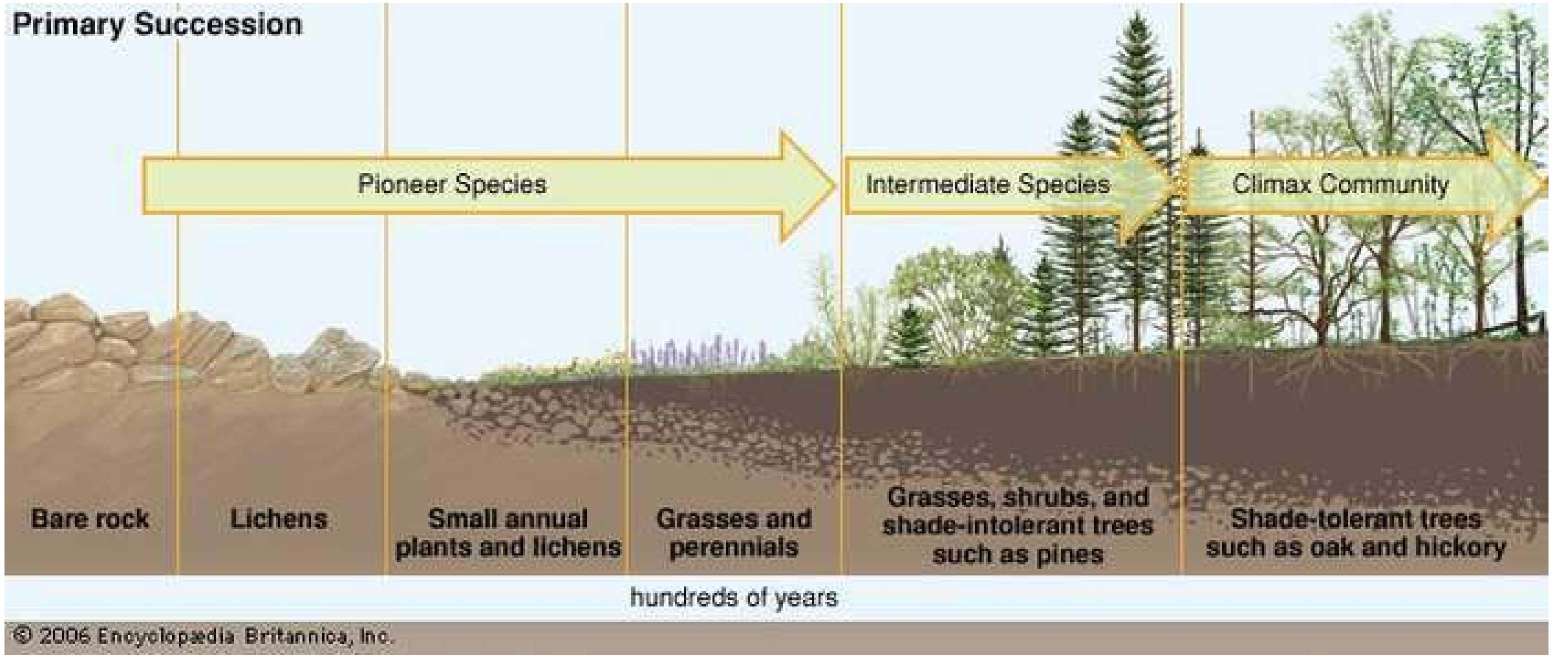


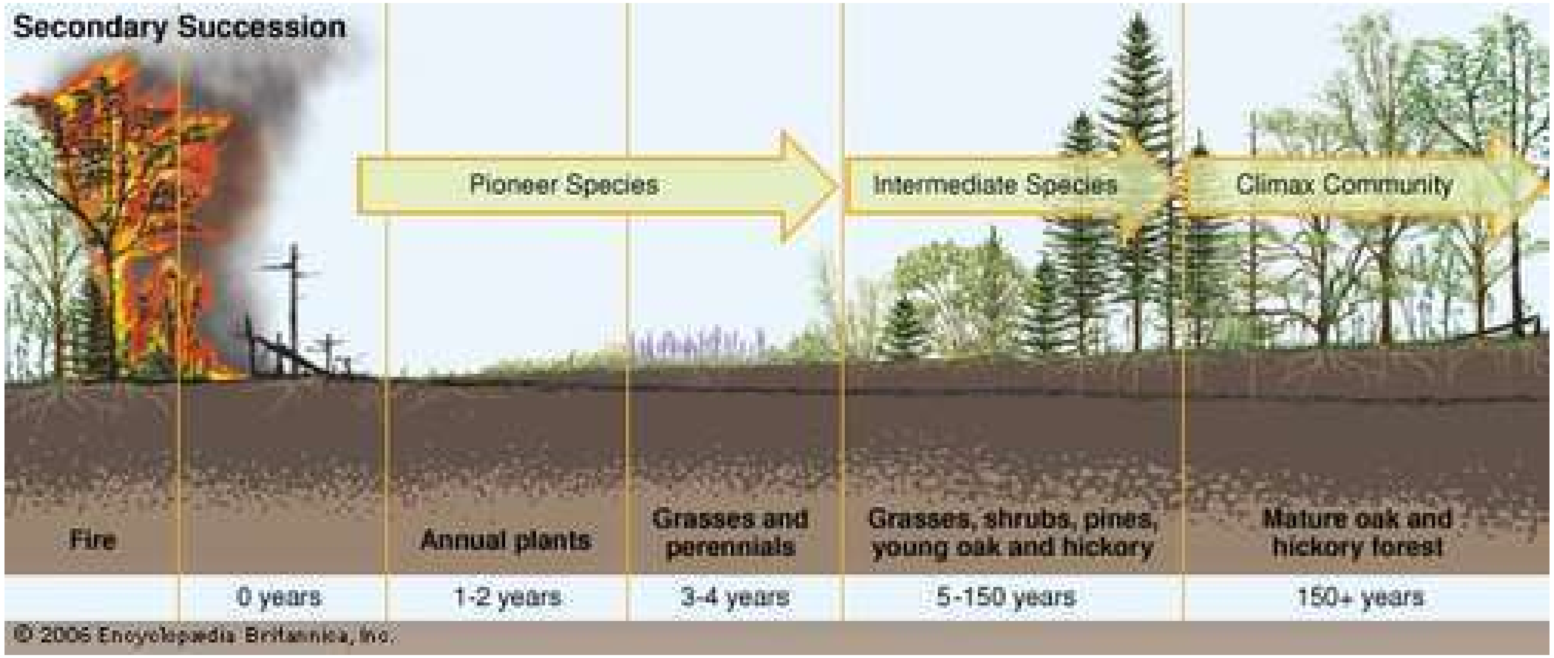
Disturbances

- Minor disturbances include localized wind events, droughts, floods, small wildland fires, and disease outbreaks in plant and animal populations.

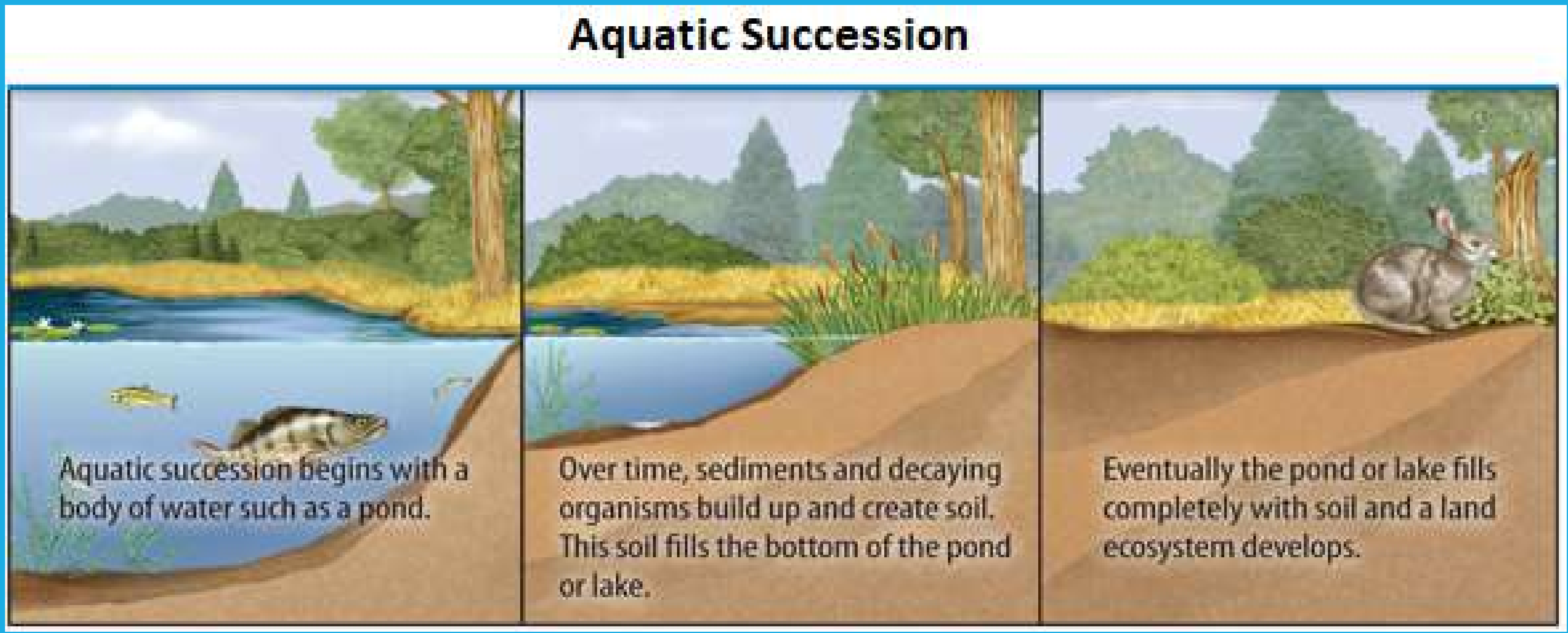
- In contrast, major disturbances include large-scale wind events (such as tropical cyclones), volcanic eruptions, tsunamis, intense forest fires, epidemics, ocean temperature changes stemming from El Niño events or other climate phenomena, and pollution and land-use.







Secondary succession looks very similar to primary succession but does not require soil forming pioneer species.


Aquatic Succession
Aquatic succession begins with a body of water such as a pond.
Over time, sediments and decaying organisms build up and create soil. This soil fills the bottom of the pond or lake.
Eventually the pond or lake fills completely with soil and a land ecosystem develops.

Ecological succession is a natural process. But, almost every human intervention from clear cutting and overhunting to farming and pollution can accelerate ecological succession.

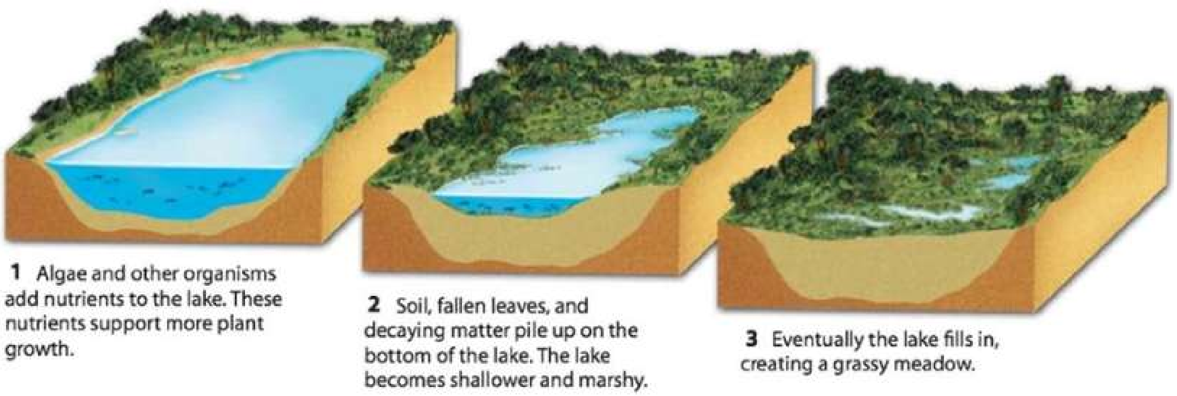
For example, today the rate of aquatic succession is sped up by the following disturbances:

❖ Some organic matter and nutrients (nitrogen and phosphorus) from sewage treatment plants enter streams and lakes.

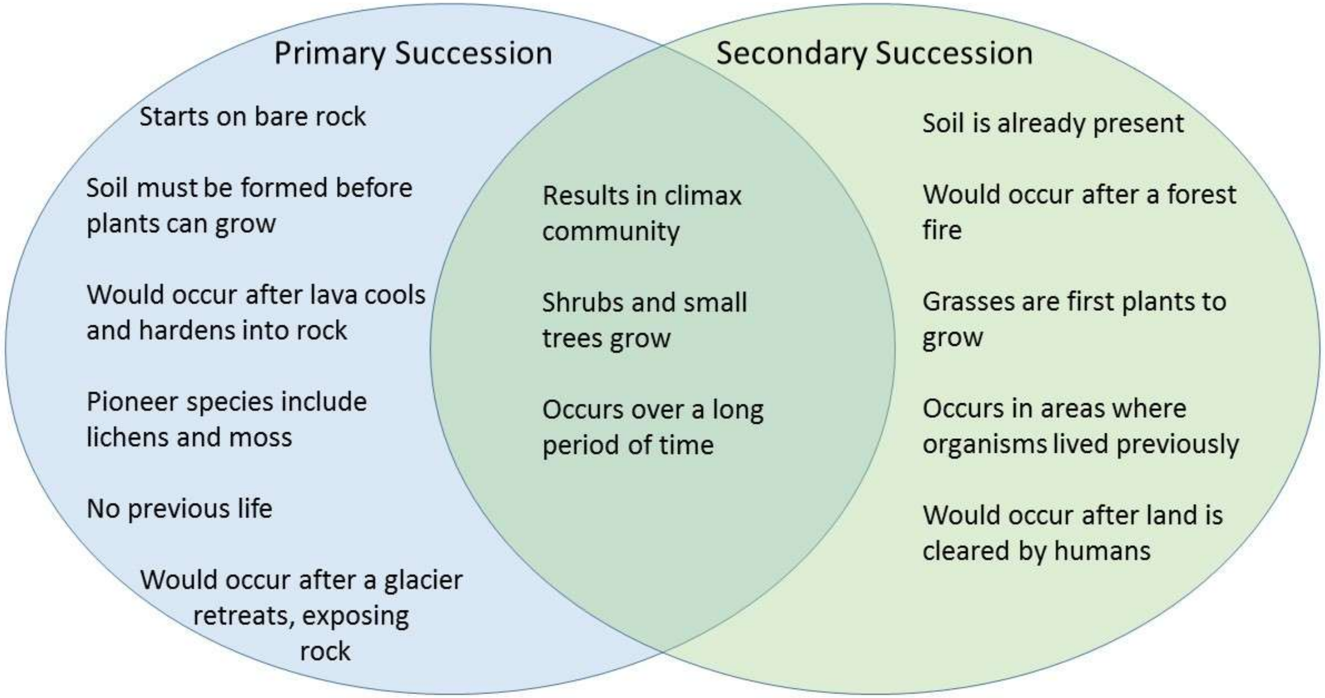
❖ Fertilizers containing nitrogen and phosphorus can run off into water bodies.

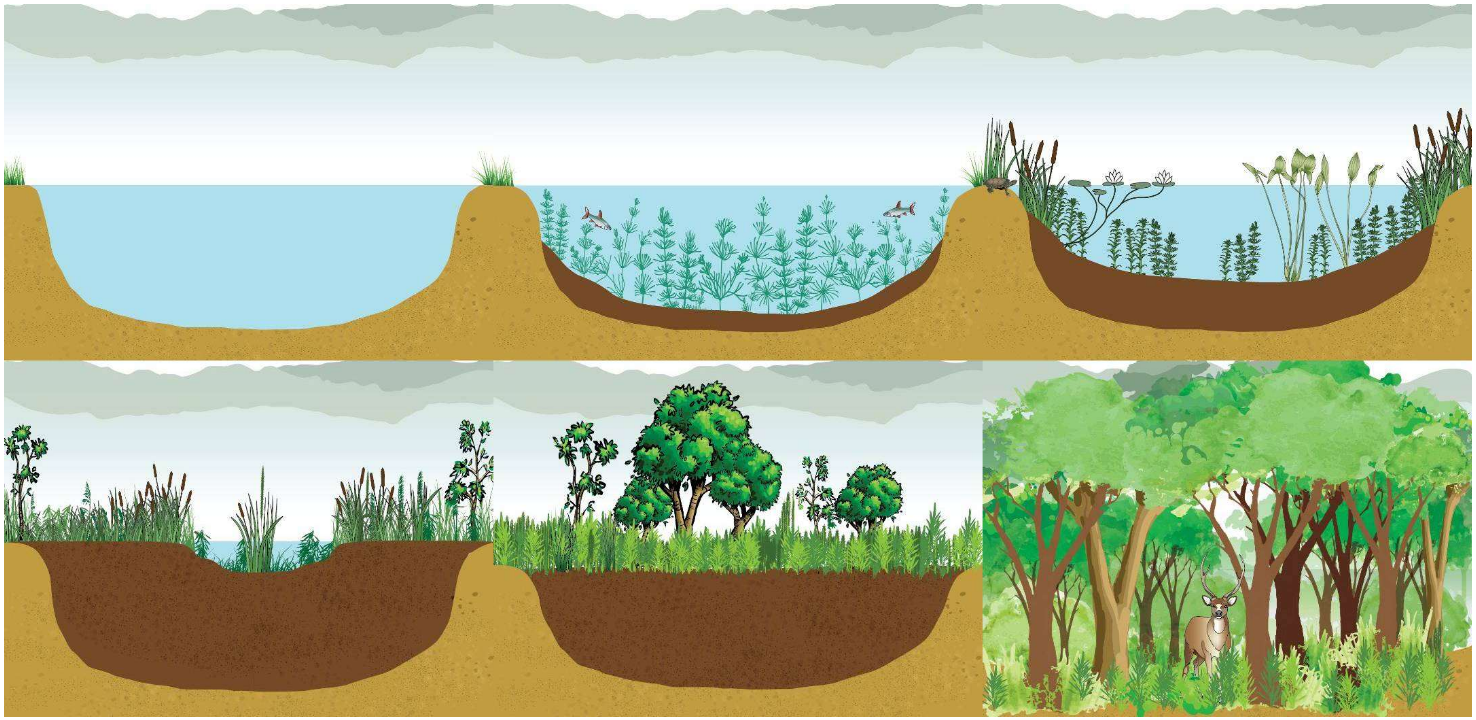
❖ Animal wastes also may add organic nutrients to water.

Small lakes and lakes near human settlements undergo aquatic succession more rapidly.

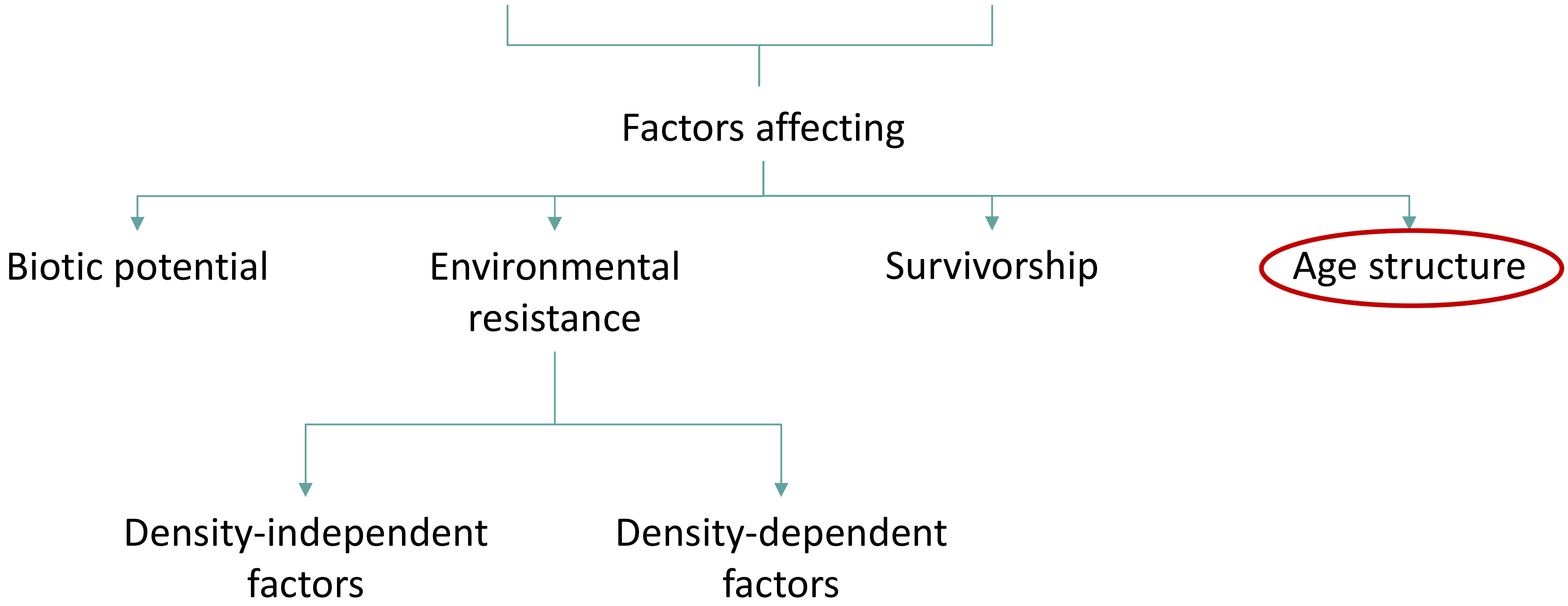


Venn Diagram Answer Key

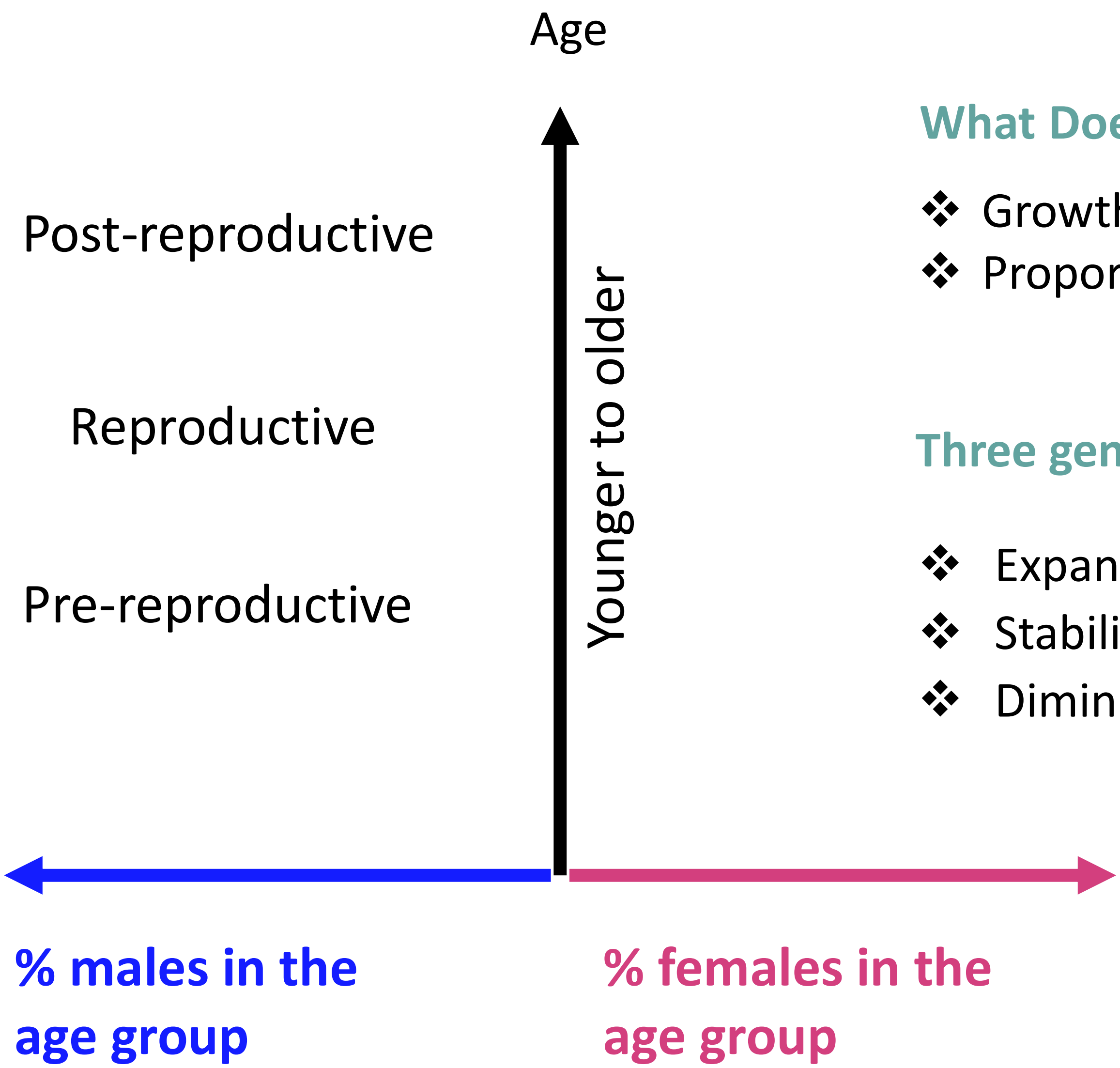




Population Growth



Age Structure Diagrams



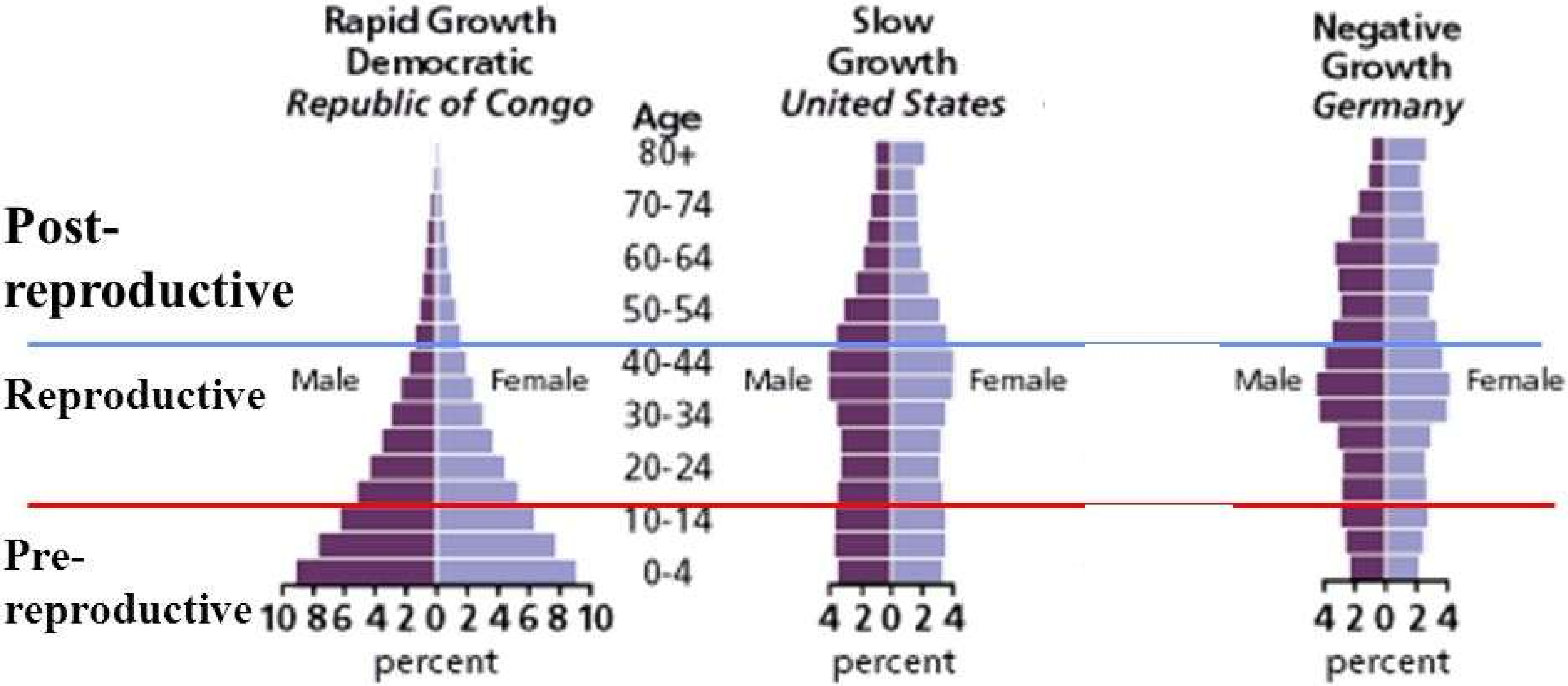
What Does the Age Structure Diagram Indicate?

- Growth Patterns
- Proportional Distribution in Age Categories

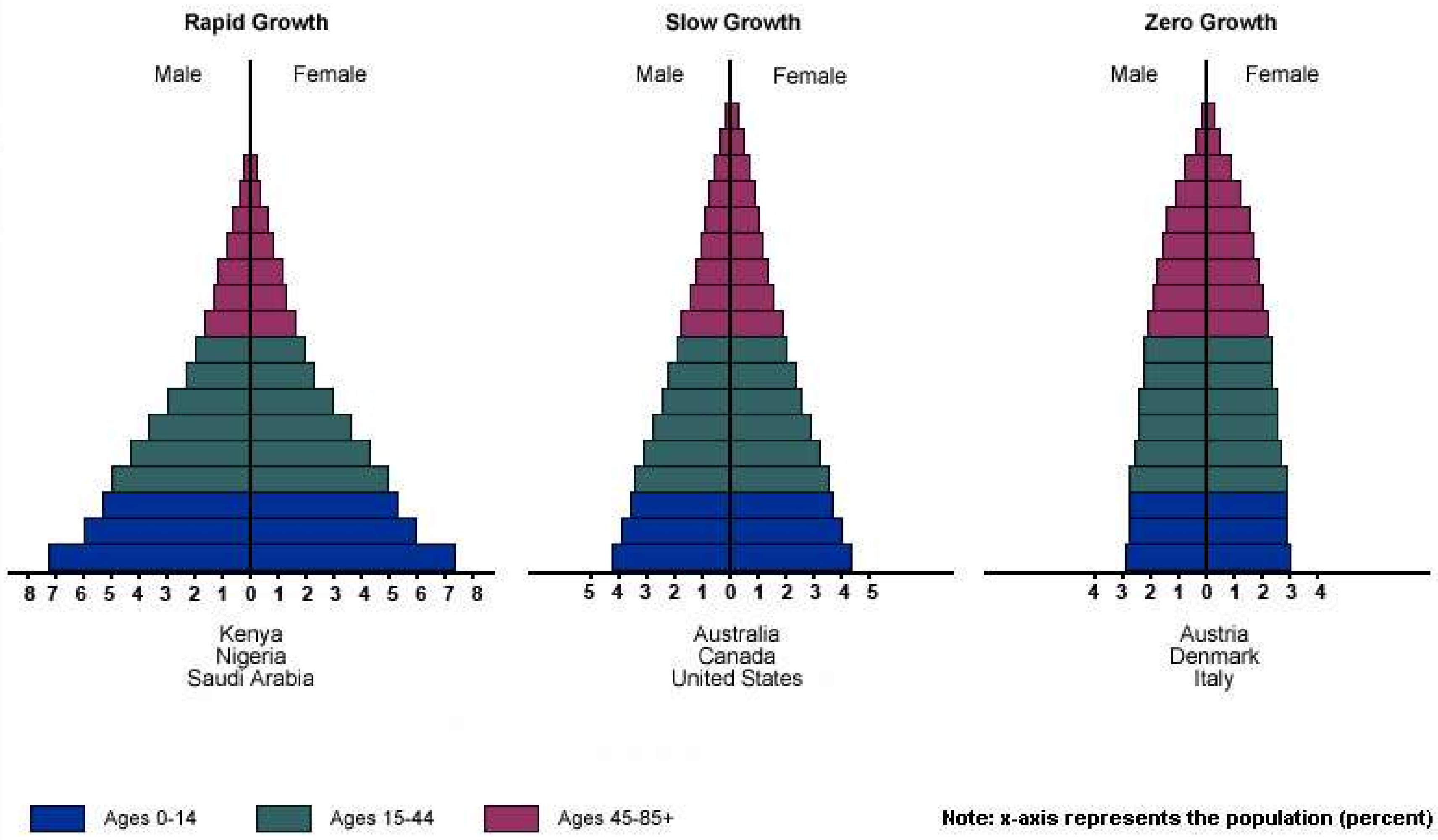
Three general types of age structure diagrams:

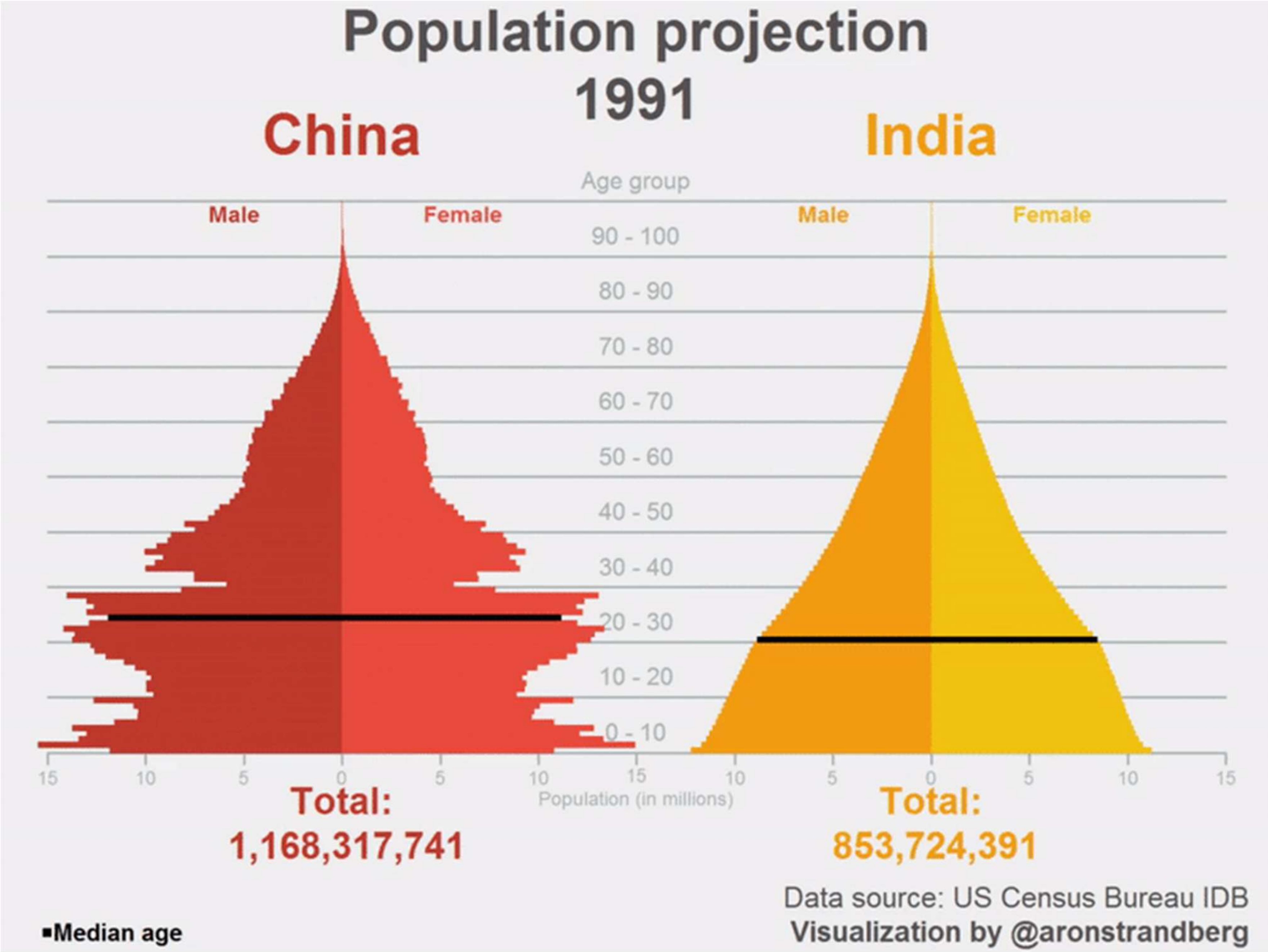
- Expanding
- Stabilizing
- Diminishing

Expanding Growth 2.7% | **Stabilizing 0.6%** | **Diminishing –0.2%**



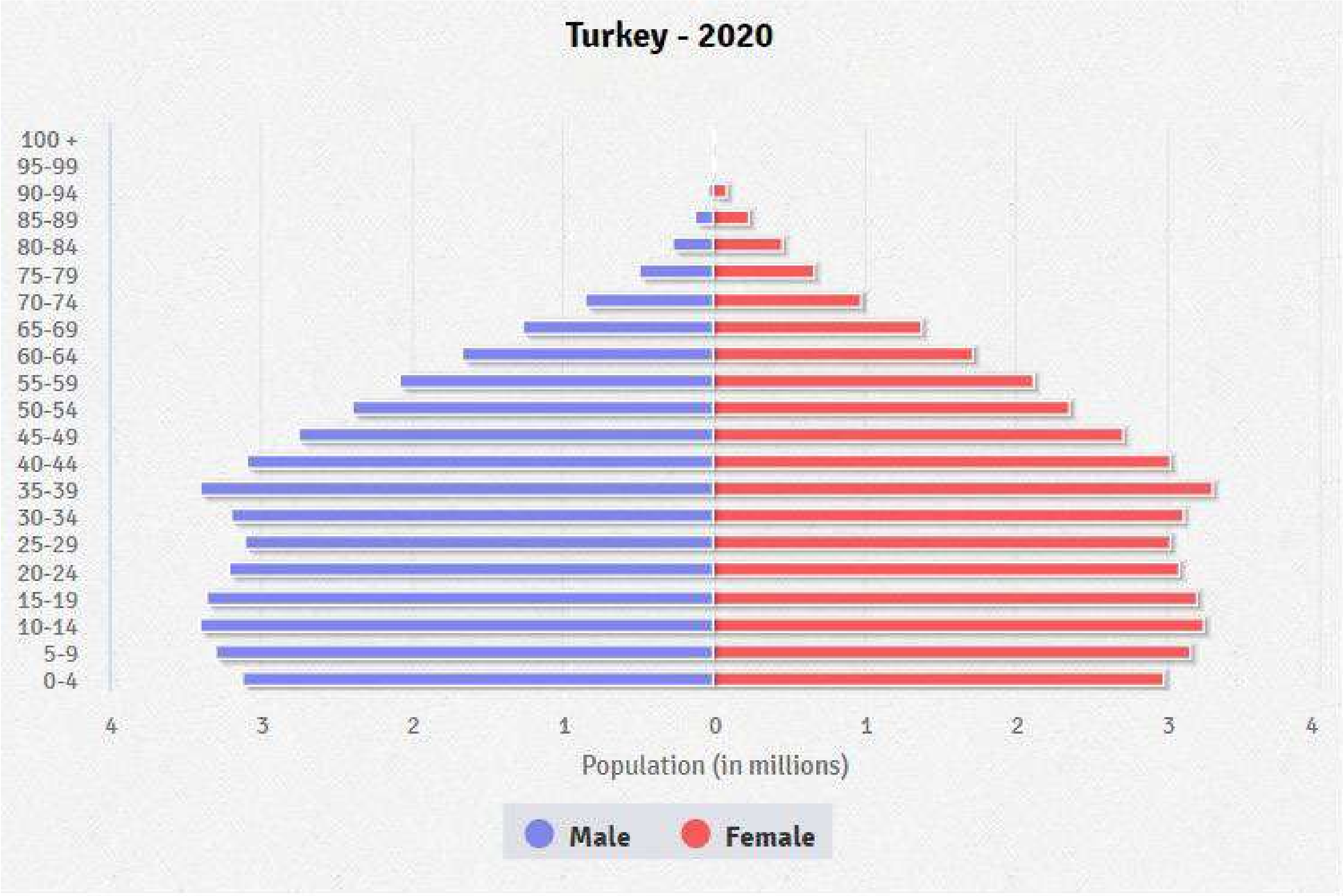
The age structure of a population is the proportion of individuals in different age-groups





Change in world population:
https://www.youtube.com/watch?v=AYLTeZz010A

Age Structure in Turkey: The Turkish Population is biased towards young people.

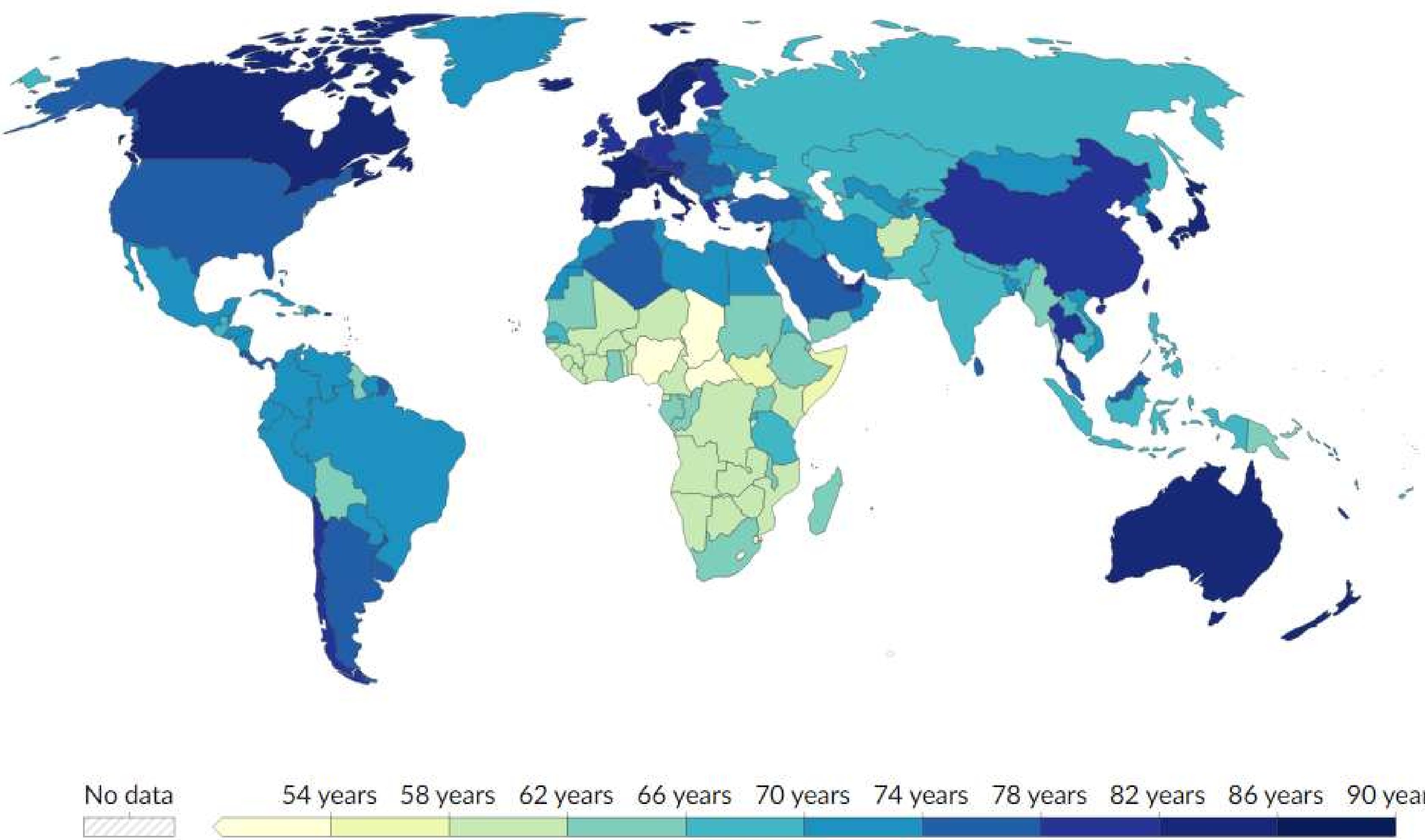


0-14 years: 23.41% (male 9,823,553/female 9,378,767)
15-24 years: 15.67% (male 6,564,263/female 6,286,615)
25-54 years: 43.31% (male 17,987,103/female 17,536,957)
55-64 years: 9.25% (male 3,764,878/female 3,822,946)
65 years and over: 8.35% (male 3,070,258/female 3,782,174) (2020 est.)

https://www.indexmundi.com/turkey/age_structure.html/

https://tr.pinterest.com/pin/657947826781330450/

Life expectancy, 2021



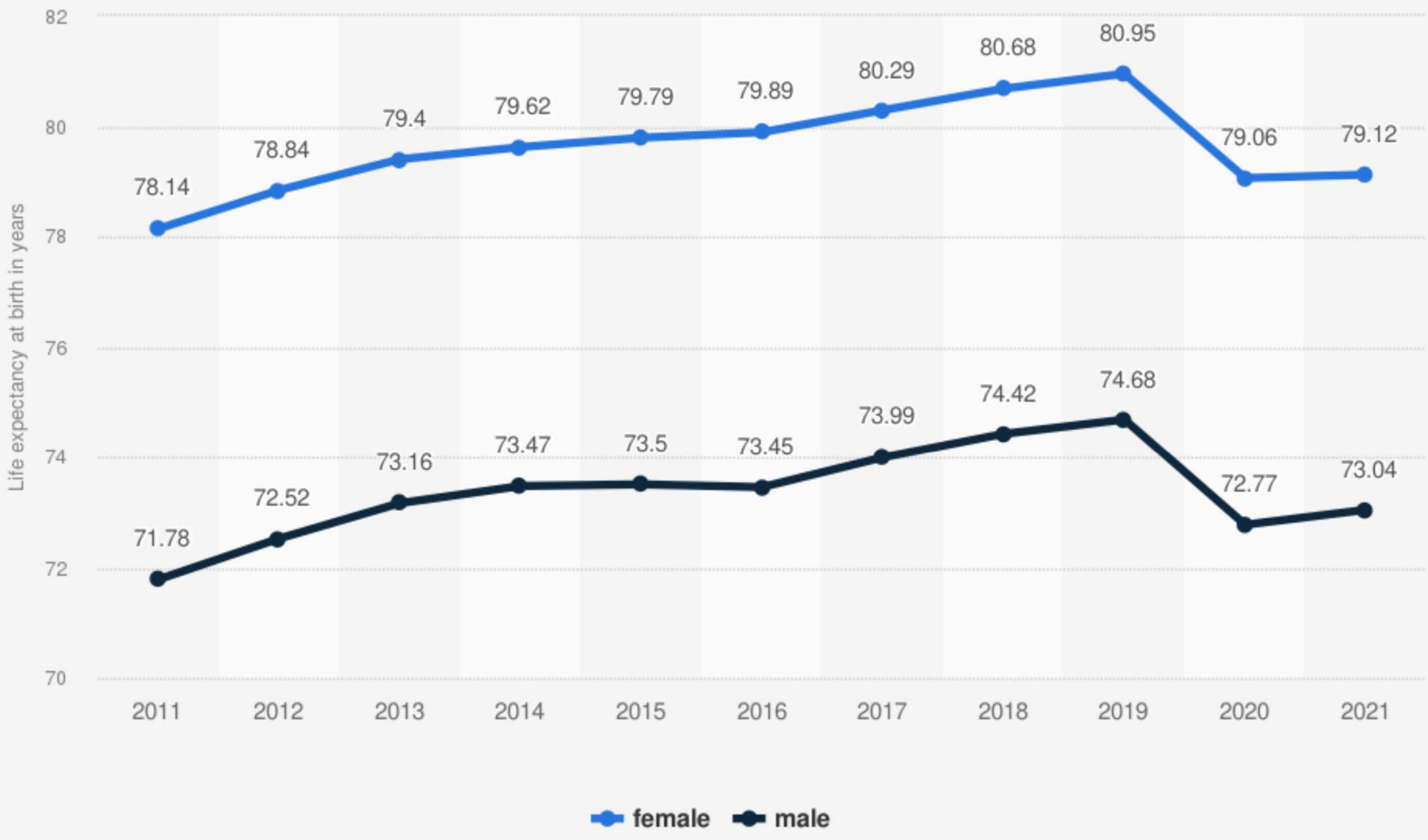
Source: UN WPP (2022); Zijdeman et al. (2015); Riley (2005)

OurWorldInData.org/life-expectancy •

Note: Shown is the 'period life expectancy'. This is the average number of years a newborn would live if age-specific mortality rates in the current year were to stay the same throughout its life.

Change in life expectancy:
https://www.youtube.com/watch?v=M45dnFfn7go

Turkey: Life expectancy at birth from 2011 to 2021, by gender



Source
World Bank

Additional Information:
Turkey